# From Pilot toward Conditional Scale: A Stage-Gated AI4SE Path for Mid-to-Large Software Organizations

Juan Qiu*, Qilong Yan*
*Inspire Group, China
Emails: juan.qiu@inspiregroup.com, qlyan@inspiregroup.com

***Abstract*—Mid-to-large software organizations—multi-site engineering groups with enough process inertia and cross-team handoff that a tool trial does not diffuse by itself—often equate an AI-for-software-engineering (AI4SE) assistant rollout, or a completed pilot increment, with readiness to scale. As generation cost falls, scarce capacity shifts from coding throughput to *definition and verification*—knowing what is correct and proving it. We report a stage-gated transformation path for that setting—Explore, Consolidation, and conditional Scale—in which each gate is an evidence review rather than a calendar event. At Org-A, a six-week Explore engagement produced a citable method, countable opportunity assets, and a readiness-labeled executable-asset ledger; management passed the Explore gate and opened Consolidation, but did not admit Scale. As a secondary recognition observation, Org-B—a manufacturing enterprise whose incumbent process is Integrated Product Development (IPD)—selected the same path after co-located alternatives were judged infeasible and is now in a mid-July to mid-September 2026 Explore that remains open. The paper contributes the path and the Org-A gate-and-asset record; Org-B supplies only a decision-fact vignette, not a second completed case. It does not claim organization-wide scale, completed Scale admission, or platform-measured return.**

***Index Terms*—AI for software engineering, stage-gated transformation, industrial experience report**

## I. Introduction

Sponsors of AI4SE programs face a recurring management error: equating access to coding assistants, or completion of a facilitated pilot, with a scale-ready organizational capability. The first produces local tool use; the second may produce a one-off demonstration. Neither, by itself, answers whether a new team can run a complete cycle under agreed guardrails. Guidance on developer productivity already warns that a single activity or throughput slogan is a poor decision aid [1]. A sharper industrial reading is that AI coding assistance compresses *implementation* cost while amplifying errors that originate in vague intent, weak specifications, and unverified agent output: the bottleneck moves toward *definition scarcity* and verification capacity. The corresponding transformation error is to treat a well-received pilot as authorization for organization-wide diffusion.

We report a stage-gated path for advancing industrial AI4SE without treating calendar completion as admission—validated primarily by Org-A's real gate disposition and asset ledger, not by a method brochure alone. A stage has a purpose, work products, and an observable exit that is also an admission test for the next stage. Explore asks whether a method can be exercised end-to-end in a real delivery context. Consolidation asks whether the pilot team can repeat and teach that method. Scale, entered only conditionally, asks whether a new team can complete a cycle independently. Org-A supplies the primary longitudinal evidence; Org-B supplies only a bounded recognition vignette (local adoption after infeasible alternatives, Explore still open). The transferable object is the path-plus-gate contract—not a tool catalog and not a completed rollout.

We organize the experience around three questions:

- **Q1.** How can an industrial AI4SE path from pilot toward scale be designed (goals, deliverables, and gate or exit tests)?
- **Q2.** What assets, signals, and admission boundaries did Org-A's completed Explore produce in support of Consolidation and of only-conditional Scale?
- **Q3.** While Org-B is still in Explore, what does a local adoption decision after infeasible alternatives show—and what does it not show?

**Methodological stance.** This manuscript is an industrial experience report [2]. Facilitators co-designed the path with client teams, but the claims we defend are those Org-A management actually gated (Explore pass / Scale hold) and the readiness labels in the handoff ledger—practitioner-validated decisions, not a consultant self-score. Org-B remains a secondary recognition vignette [3]. We do not claim multi-case theoretical saturation, a completed design-science evaluate cycle, researcher-owned action-research agency, or completed Scale at either organization.

**Contributions.** (1) a stage-gated AI4SE path with an admission contract (citable / runnable / transferable readiness; stub $\neq$ ready); (2) Org-A evidence that Explore completed, Consolidation started, and Scale admission was not met—stated as a three-layer feasibility argument rather than as assumed diffusion; (3) a bounded Org-B decision fact only:

J. Qiu is Principal AI4SE Consultant at Inspire Group (Ph.D., Software Engineering, Tongji University; 18+ years in software engineering, R&D productivity, and enterprise architecture). Q. Yan is AI4SE Engineering Advisor at Inspire Group (M.S., Computer Science, Sichuan University; 10+ years in enterprise architecture, DevOps, and AI-agent engineering). Inspire Group is a technology-services firm formed from the former Thoughtworks China business after its acquisition by Hillhouse Capital (announced 2025); the Inspire authors facilitated the Org-A and Org-B engagements. Industrial partners remain anonymized (Org-A, Org-B); no client co-authors.

the same path was selected after alternative approaches were judged infeasible, under an IPD constraint, with Explore still open (not a second completed gate packet).

Bounded Explore-phase effectiveness and maturity signals appear here only as gate inputs with stated limits. A companion measurement manuscript (separate contribution; dual-lens labeled reporting of capability and delivery signals at Org-A) owns how those signals are stratified and later promoted from estimate to platform measurement; we do not reproduce its tables or claim concurrent review of that companion manuscript here.[1]

Section II supplies the conceptual overlay. Section III defines the stages and gates. Section IV reports Org-A. Section V reports Org-B. Sections VI–VIII discuss takeaways, threats, related work, and the claim ceiling.

## II. Conceptual Background

The stage-gated path is the contribution of this paper. This section supplies a compact vocabulary for *why the path is staged* and *what it carries*. It is not a second framework to be evaluated on its own, and it is not a restatement of a practitioner slide deck.

### A. Domain Fit: Why Exploration Is Not Instant Scale

Snowden and Boone's Cynefin framing distinguishes decision domains in which best practice, expert analysis, or probe-and-sense behavior is appropriate [4]. Early industrial AI4SE work is typically *complex*: workable method is emergent, so Explore is a probe that must produce inspectable method assets rather than a purchased best-practice pack. Consolidation moves durable findings toward the *complicated* domain: playbooks, harness hardening, and seed coaches support sense–analyze–respond transfer. Scale entry is admitted only when enough of the method is stable to diffuse without re-running full discovery—closer to clear or well-analyzed complicated work, never by calendar alone. The gates are therefore domain-fit devices: treating a Complex probe as Clear rollout is the failure mode this paper records as "pilot success without Scale admission."

### B. Industry Fault Lines as Staging Rationale

Senior-practitioner syntheses from Thoughtworks' Utah and Europe Future of Software Engineering retreats highlight fault lines that motivate staging [5], [6]: engineering rigor migrating into specifications, tests, and constraints; a supervisory *middle loop* between coding and delivery; verification (not generation) as scarce capacity; harness ownership around the agent; and a two-clock mismatch in which code-production time collapses while decision time does not. We use that map only as industrial *rationale*. Explore must freeze probe evidence; Consolidation must harden verification, harness ownership, and seed coaches; Scale is admitted only when decision and verification capacity can absorb a receiving team. Retreat anecdotes are not Org-A/Org-B facts, and retreat consensus is not empirical validation of our gates; denser positioning appears in Section VIII.

[1]Companion status at freeze: authoring for a peer-reviewed software-engineering venue; unit of analysis = measurement scheme, not the stage-gated path. No Org-A workshop or maturity numbers from that manuscript are imported here.

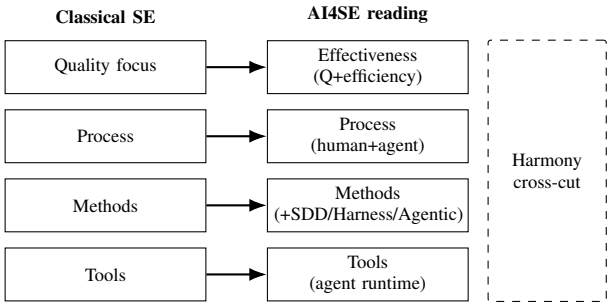


Fig. 1. Inherited layered software-engineering structure with upgraded AI4SE semantics (conceptual). Harness/SDD/Agentic sit primarily in Methods; Harmony cross-cuts rather than stacking a fifth layer.

### C. Layer Inheritance: Pressman Structure, Upgraded Semantics

Classical software-engineering textbooks present a layered view in which a quality focus grounds process, methods, and tools, and in which more visible layers depend on less visible ones [7]. Figure 1 states our industrial reading: the *structure* is inherited; the *semantics* must be upgraded for human–agent delivery. Quality focus becomes *Effectiveness* (quality plus waste-aware efficiency). Process, methods, and tools retain their roles, but executors are humans *and* agents. Harness Engineering, Spec-Driven Development (SDD), and Agentic Engineering are placed in the *methods* family (discipline and assets), not as a replacement for the process layer or as a tools-only purchase. *Harmony* is a cross-cutting human–agent contract, not a fifth stacked layer.

### D. Effectiveness, Harmony, and Four In-Path Directions

We use two cross-cutting concerns—*Effectiveness* and *Harmony*—and four capability directions: SDD, Agentic Engineering, Harness Engineering, and an AI4SE Operating Model. Figure 2 maps them onto the stages of Section III. We do not treat these labels as a maturity model, a vendor stack, or a substitute for the gates.

Effectiveness is the requirement that each stage produce inspectable evidence that the selected work actually ran under local constraints—a citable method, a validation-vehicle run-through, or a new-team cycle—rather than attendance counts or tool-license counts. Harmony is the requirement that human and agent accountability be *operationalized*, not merely named. In the engagements we distinguish four complementary functions that may be held by fewer people in a small team but must not be vacant: *intent ownership* (what to build and what success means), *specification stewardship* (living specs and change deltas as the shared source of truth), *AI-asset orchestration* (skills, recipes, and harness units that make the method executable), and *verification leadership* (acceptance and gate sign-off). A working rule is *contract separation*:

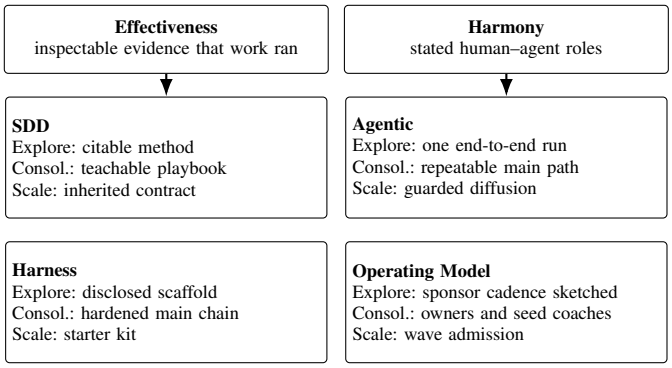


Fig. 2. Effectiveness, Harmony, and four capability directions mapped onto the stage-gated path. The overlay is conceptual; it is not a second evaluated contribution.

the party that proposes or applies a change must not be the sole party that verifies it—agents may draft and execute, but humans retain intent and verification authority. Both Effectiveness and Harmony are inputs to G1 and G2. They are not outcome multipliers, and this paper does not report them as measured organizational return.

The four directions are stage-relative obligations.

**SDD** converts intent into specifications that agents and humans can execute, review, and accept. Explore must produce a first citable method; Consolidation must make that method teachable; Scale uses it as the contract a receiving team inherits. Spec-driven development paired with agentic execution is an active industrial theme; Gupta discusses the pairing in a preprint on high-accuracy agentic frameworks [8]. We cite that work as thematic context, not as an evaluation of Org-A or Org-B.

**Agentic Engineering** turns those specifications into composable, reviewable agent work—skills, commands, and recipes—rather than ad-hoc prompting. Explore needs one end-to-end agentic run; Consolidation needs the principal path to be repeatable by the pilot team; Scale admits that path only with declared guardrails.

**Harness Engineering** is the runtime—context, tools, permissions, feedback, and audit—in which agents operate. Explore may leave the harness as a disclosed scaffold; Consolidation must harden the main chain; Scale requires a starter kit another team can run. A stub is not a production harness; Section IV operationalizes that distinction in the Org-A ledger.

**Operating Model** covers owners, coaching, sponsor cadence, and wave admission—including named holders of the Harmony functions above. It is sketched in Explore, staffed in Consolidation (named owners and seed coaches), and enforced at Scale entry. Without it, a hardened harness still has no authorized receiving team.

### *E. Certainty Boundary: User Harness, Model, and Method Assets*

Industrial harness writing emphasizes bounded contexts among users, coding agents, and models [9]. Figure 3 is our AI4SE reading of that idea for transformation work: organizational certainty does not live in the model alone. We

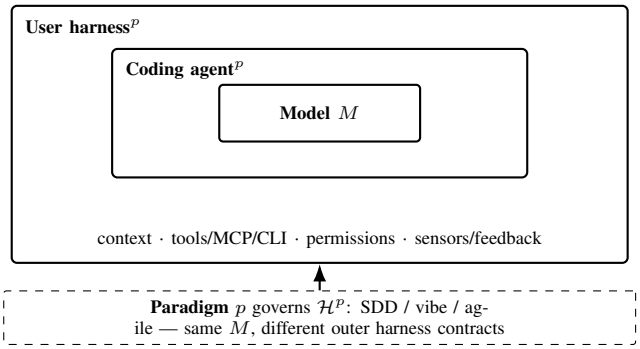


Fig. 3. Certainty boundary for AI4SE enablement (conceptual adaptation of harness bounded-context thinking [9]). Top: nested layers realizing (1). Bottom: paradigm $p$ selects the outer contracts (Eq. (2)); method assets feed those outer units. Agentic engineering composes them—it is not a competing fourth paradigm.

write the staged enablement object as a paradigm-conditioned composition

$$\mathcal{H}^p(x) = \mathrm{UserHarness}^p\big(\mathrm{CodingAgent}^p\big(\mathrm{Model}(x)\big)\big), \quad (1)$$

where $p$ is the governing delivery paradigm (for example spec-driven development versus loosely guided "vibe coding," or an agile variant). The inner $\mathrm{Model}$ is comparatively stable across engagements; the outer two layers are where Explore must freeze inspectable structure and Consolidation must harden ownership. The same model under different $p$ therefore yields different $\mathcal{H}^p$: an SDD-oriented user harness privileges contracts, acceptance checks, and reviewable agent recipes, whereas a vibe-oriented harness privileges fast prompt–edit loops and lighter formal gates. Neither is universally "correct"; the gate packet must record which $p$ was chosen and what evidence that choice requires.

Operationally, each outer layer is itself a composition of smaller units around a frozen model operator $M$:

$$\mathcal{H}_{\mathrm{single}}(x) = h_k\big(\cdots h_1\big(M(h_0(x))\big)\cdots\big), \quad (2)$$

with $h_0$ assembling context and guides, $h_1$ parsing and schema checks, and later $h_i$ covering tools, permissions, sensors, and feedback. *Agentic engineering* is not a fourth paradigm competing with SDD or vibe coding; it is the engineering style that turns a chosen $p$ into composable, executable, and verifiable AI assets—skills, subagents, commands, rules, and hooks—inside $\mathrm{UserHarness}^p$ and $\mathrm{CodingAgent}^p$. Method assets (playbooks, SDD contracts, starter kits) feed those outer layers; they do not replace the model. The harness must remain an open class across SDD stacks (for example OpenSpec-, Superpowers-, or Spec-Kit-style frameworks, and future peers), not a lock-in to one template—a constraint Org-B makes explicit in Section V.

Skipping Consolidation typically leaves SDD prose and a stubbed harness in the hands of a new team—the failure mode Section III names as equating a pilot with scale. The conceptual layer therefore explains *why* the gates ask for method, asset, ownership, and transfer evidence; it does not add a parallel transformation method.

## III. Stage-Gated Path

### A. Why a Gate, Rather Than a Calendar, Advances the Path

We define an AI4SE transformation path as a sequence of capability-building stages separated by explicit management decisions. A stage has a purpose, work products, and an observable exit condition. Its exit condition is also an admission test for the next stage: elapsed time can trigger a review, but cannot substitute for evidence. This distinction is important because a pilot can finish on schedule while leaving its method tacit, its automation fragile, or its ownership unresolved. Collapsing code-production time does not by itself collapse decision and verification time—another reason gates must review evidence rather than calendars.

The transformation path nests two control loops. *Stage gates* (G1/G2) decide whether the organization may enter Consolidation or admit a Scale wave. Inside delivery work, shorter *change gates*—align a delta specification, verify a candidate increment, then merge and archive—protect each change under the chosen paradigm $p$. Stage gates ask whether method, assets, and ownership are transferable; change gates ask whether a particular change is definition-complete and evidence-backed. Confusing the two is a common failure: a team that passes local change reviews can still fail G2 if receiving teams cannot inherit the method.

The path instantiated at Org-A comprises *Explore*, *Consolidation*, and *Scale* (Fig. 4). Explore asks whether the method can be exercised end-to-end in a real delivery context. Consolidation asks whether the resulting method can be repeated and taught by the pilot team. Scale, which is entered only conditionally, asks whether a new team can run a complete cycle independently while retaining agreed guardrails. Outputs therefore flow forward as dependencies: the first-generation playbook and pilot assets feed hardening; the hardened method, executable starter kit, role owners, and seed coaches then support controlled diffusion.

### B. Stages, Deliverables, and Exit Criteria

**Stage 1—Explore.** The purpose of Explore is feasibility, not broad deployment. In the Org-A instantiation, the indicative duration was approximately six weeks, spanning diagnosis and scenario selection, four workshop weeks over twelve opportunity themes, co-creation of an initial playbook and executable-asset scaffold, and use of a pilot change request as a validation vehicle. The exit criterion was an end-to-end run-through of the method chain. It was deliberately not completion of every feature in the validation vehicle.

**Stage 2—Consolidation.** Consolidation is the bridge between one successful run and a transferable organizational capability. Its indicative Org-A window was one to two months. The work comprises revising clauses that read correctly but do not run in practice, hardening the principal executable path from stubs to usable assets, assigning role ownership, and rehearsing coach-the-coach transfer. Its intended products are a standardized playbook v2, a hardened toolset, and seed coaches. Hardening follows two practical rules drawn from the enablement practice: *do not promote immature automation* (skills and recipes must stabilize before they are wrapped as autonomous agents), and *keep apply separate from verify* (an execution agent must not also issue the verification verdict). The exit condition is "team can teach": pilot-team members can explain and transfer the method, rather than only reproduce a consultant-led demonstration.

**Stage 3—Scale.** Scale is quality-consistent, wave-based diffusion rather than simultaneous organization-wide rollout. Admission requires the Consolidation exit, named role owners, a starter kit for sibling teams, and reserved sponsor cadence and resources. Seed coaches then enable successive teams and review practice consistency. The terminal evidence is a fresh team's independent completion of a cycle under the guardrails. Thus, a calendar target can prompt a G2 review, but an unmet G2 blocks admission.

### C. Gate Semantics

G1 is a *method–asset gate*. Management asks whether a playbook can be shown and cited with its gaps visible; whether opportunity-point specifications and best practices can be inventoried; whether executable assets are distinguished as ready or stub; whether effectiveness and maturity signals carry their measurement boundaries; and whether the Consolidation-to-Scale path is concrete enough to discuss. A positive G1 therefore licenses method hardening, not broad rollout. Table I makes this decision contract explicit.

G2 is stricter because it concerns transfer beyond the pilot setting. It tests whether know-how has moved from individuals into teachable guidance, executable assets, role ownership, and an enablement mechanism—and whether verification authority remains human-held under the nested change gates. The resulting admission decision is reversible and wave-specific: if a new team cannot operate with the starter kit and bounded coaching, the organization returns the deficient asset or role arrangement to Consolidation. This interpretation makes a gate a learning control, rather than a ceremonial approval checkpoint.

### D. Operating a Gate as an Evidence Review

A usable gate needs a review packet, a decision owner, and a recorded disposition. The review packet should present the stage goal first and then map each promised output to an inspectable artifact or observation. A readiness label accompanies each item so that "exists," "runs on the primary path," and "transferable to a new team" cannot collapse into one category. Known gaps remain in the packet as decision inputs. Removing them from view would make an approval easier to obtain but less useful for planning the next stage.

The decision owner evaluates sufficiency for the *next* stage, not perfection in the current one. At G1, an incomplete executable scaffold can be acceptable when it is disclosed and when the Consolidation plan names the hardening work. At G2, the same scaffold is a blocker because a receiving team would depend on it. This relative interpretation prevents two common errors: rejecting useful Explore evidence because

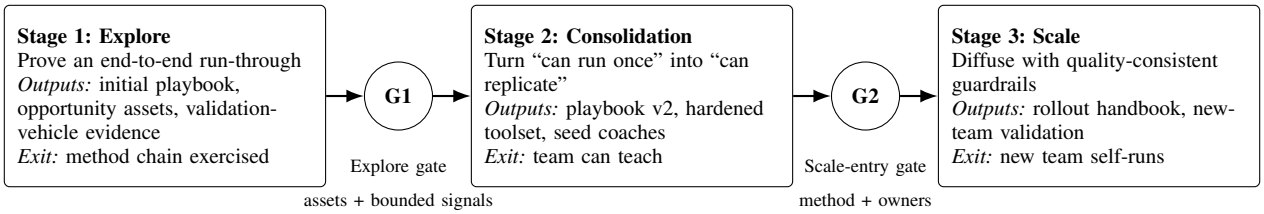


Fig. 4. Stage-gated AI4SE transformation path. The figure specifies evidence required to advance; it does not indicate that Org-A completed Scale.

TABLE I
DECISION TESTS AT THE TWO STAGE BOUNDARIES

| Boundary | Evidence reviewed | Positive decision | What the decision does not imply |
|---|---|---|---|
| G1: Explore close-out | Citable playbook with gaps stated; inventory of converged and observed opportunity assets; readiness of executable assets stated as ready or stub; bounded effectiveness/maturity signals; discussable next-stage path | The method chain ran end-to-end and the organization may enter Consolidation | All validation-vehicle features are complete; executable assets are ready for broad use; measured organization-wide return has been established |
| G2: Scale entry | Team-can-teach evidence; hardened main execution path; named role owners; starter kit; sponsor rhythm and resources | A controlled wave may admit a new team with seed-coach support | The Explore pilot alone authorizes diffusion; every team is already independent; rollout outcomes are known in advance |

every asset is not yet finished, and admitting Scale because unfinished assets were tolerated at Explore closeout.

A gate disposition has three possible forms. *Advance* means the exit condition is met and the next stage may start under its declared scope. *Advance with conditions* records a narrow action whose completion is checked before a dependent activity begins; it must not be used to waive a core exit condition. *Hold* returns named gaps to the current stage. In all three forms, the evidence and rationale remain available for the next review. This creates continuity across decisions and makes negative evidence—for example, a stubbed primary path or missing owner—actionable rather than embarrassing.

The gate also separates two kinds of uncertainty. Product uncertainty asks whether AI4SE can improve the selected work under local constraints; Explore reduces it through actual method use. Transfer uncertainty asks whether another team can reproduce that use without recreating the method; only Consolidation and a new-team test can reduce it. Treating these uncertainties separately explains why a positive G1 and a negative G2 can be the expected, internally consistent outcome of the same review.

### *E. Three-Layer Scale-Feasibility Argument*

The path permits a useful feasibility judgment before claiming completed diffusion. We structure that judgment in three layers.

**Layer 1—the path is designed.** Goals, work products, exits, and admissions are stated before scale readiness is assessed. This makes the claim falsifiable: missing ownership, hardening, or transfer evidence can stop advancement even when a pilot was well received.

**Layer 2—Explore is validated end-to-end.** For Org-A, the initial playbook, opportunity assets, and validation vehicle exercised the method chain under real delivery constraints. The evidence supports "can run once" and justified passing G1; it does not establish repeatability across teams.

**Layer 3—Scale remains conditional.** Org-A's pilot succeeded at Explore, but G2 was not met at closeout. Consolidation—including asset hardening and team self-solve capability—is the prerequisite for a later Scale decision. The feasibility judgment therefore rests on an explicit path, observed end-to-end use, a readiness-labeled asset ledger, and testable admission conditions. It does not rest on an assumed organization-wide return or on diffusion that has not occurred.

### *F. Anti-Patterns Exposed by the Gates*

**Tool-only transformation** treats access to a coding assistant or plugin catalog as the change itself. It omits the specifications, verification, context, roles, and learning routines needed to make tool use repeatable. G1 exposes this weakness by requiring a citable method and bounded evidence; G2 exposes it by requiring transfer and ownership.

**Pilot equals scale** converts a successful demonstration directly into rollout authorization. It conflates end-to-end feasibility with cross-team replicability. Separating G1 from G2 preserves the useful result of the pilot while making the missing admission evidence visible.

**Skipping Consolidation** carries first-generation prose and stubs directly into new teams. The likely burden then moves from explicit hardening to repeated local improvisation. The Stage-2 exit—team can teach—requires the organization to remove that burden before opening a Scale wave.

**Agent self-certification** lets the same generative path that applied a change also declare the change verified. Nested change gates and the Harmony verification function exist to forbid that shortcut: drafts and execution may be agentic; gate sign-off remains human.

## IV. Case Org-A

### A. Industrial Context and Evidence Basis

Org-A is an anonymized large global container-shipping and logistics enterprise with a substantial, multi-site in-house IT organization supporting commercial and operational systems. The engagement was an approximately six-week AI4SE enablement pilot, not a scale-out program. The delivery setting used a multi-team agile model, shared requirements management, and distinct development, business-analysis, and quality roles. Quality remained a standing constraint: efficiency exploration was not allowed to relax the client's existing quality gates.

We report the case as an industrial experience and preserve a chain from each case statement to a frozen evidence slice, following the traceability principle in case-study reporting [2]. The primary case record comprises the phase timeline, the management gate definition, the L1–L4 handoff ledger, and post-closeout consolidation signals. We treat the playbook and executable assets as designed artifacts evaluated through use in their organizational environment, consistent with design-science evaluation logic [3]. This paper uses only qualitative signal boundaries from the evidence pack; measurement results are reserved for the companion measurement study.

### B. Explore Activities

The Explore phase followed a six-block arc (Fig. 5). Kickoff in late June or early July 2026 established the diagnosis, scenario lock, baseline maturity exercise, and opportunity-theme plan. Workshop W1 (early July) addressed engineering foundations: harness/context, knowledge, and brownfield specification-driven development. W2 (mid-July) covered metrics and requirements entry and included a midterm management showcase. W3 (mid-to-late July) addressed review, agent validation, and test engineering. W4 (late July) ran two tracks in parallel: playbook v0.5 authoring and specification-based validation in the delivery track. An early-August executive readout closed Explore, aligned the Consolidation path, and explicitly did not claim Scale admission.

Across those four workshop weeks, the team worked through twelve opportunity themes and co-created initial method and opportunity assets. A pilot change request supplied the validation vehicle for using the method in delivery. This choice kept acceptance focused on whether the chain from opportunity through specification and practice could be exercised, rather than on shipping all business scope. The activity record consequently supports a process-feasibility conclusion, not a claim about broad adoption.

### C. From Activities to Gate Evidence

The workshop sequence was not treated as attendance evidence. Each block was expected to leave a trace in one or more asset layers. Foundation work informed the method guidance and executable scaffold; opportunity work created specifications and best practices; delivery-track use tested whether those descriptions could guide action; and the closeout evidence pack recorded both observed signals and their limits. This trace converted a series of facilitated sessions into inspectable inputs to G1.

The parallel tracks in W4 were particularly important to that conversion. Playbook authoring alone could have produced guidance that was internally coherent but detached from delivery. Delivery work alone could have produced a one-off result whose method remained tacit. Running authoring and specification validation together created a feedback loop: delivery exposed where wording or executable support was weak, while the playbook gave the team a stable object to revise and later cite. The gate then assessed the resulting artifacts and disclosed gaps rather than inferring capability from workshop completion.

We used three readiness distinctions when interpreting the handoff. *Citable* means an artifact is stable enough to identify and discuss. *Runnable* means that its principal path works in the intended local workflow. *Transferable* adds the ability of another team to use that path with declared guardrails and bounded coaching. Explore required a citable method and evidence of local use; Scale admission would require runnable and transferable support. The distinctions explain why the L1 playbook could satisfy its G1 criterion while the largely stubbed L3 pack could not satisfy G2.

### D. L1–L4 Outcomes and Readiness

The closeout inventory was organized into four layers (Table II). The layers distinguish descriptive guidance, reusable opportunity knowledge, executable support, and evidence. More importantly, the ledger records readiness separately from existence; a catalog entry is not treated as a deployable capability.

**L1: citable method.** The frozen v1.0 handbook captured the end-to-end method—including how intent becomes living specifications, change deltas, and accepted increments—well enough to be shown and referenced at the gate. Its frozen state made the Explore result auditable; it did not mean that Consolidation revisions were complete. Playbook v2 remained a Stage-2 target. A public method card distilled for inspection (chapter map, five first principles, Living-Spec artifact chain, 5+1 stages with Align/Verify/Merge gates, paraphrased normative clauses, and declared v1 gaps) is released with the data pack;[2] the full client-facing handbook text is not.

As an illustrative L1 extract (paraphrased, not a verbatim client clause): v1.0 required that every accepted increment cite a living-specification delta id, name a human verifier distinct from the apply agent, and refuse "generated-output equals verified" shortcuts. Org-A's later Consolidation agenda explicitly included rewriting clauses that "read right but did not run" on the main L3 chain—a recurring failure mode for early playbooks. Non-ROI intensity at closeout was gate-visible without claiming return: four workshop weeks plus kickoff/readout, twelve themes (ten released), 11+11 L2 assets, a largely stubbed L3 main chain, and G1 pass / G2 hold.

**L2: countable opportunity knowledge.** The release set indexed 11 best practices and 11 corresponding specifications

[2]See `l1-method-extract.md` under the Data Availability URL.

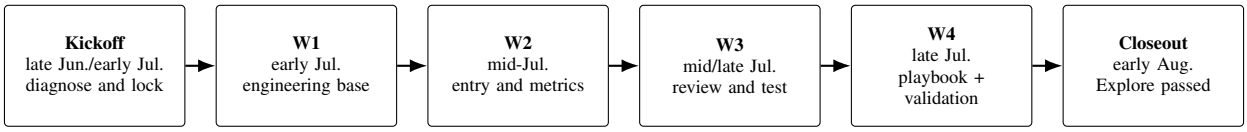


Fig. 5. Org-A Explore timeline (week-level windows). The early-August closeout passed the Explore gate and opened Consolidation; it did not pass Scale admission.

TABLE II
ORG-A EXPLORE ASSET LEDGER AT CLOSEOUT

| Layer | Asset class | Observed outcome | Readiness boundary / Consolidation need |
|---|---|---|---|
| L1 | Method playbook / operations handbook | Version 1.0 was frozen, citable, and sufficient for the Explore gate (public method card released; full handbook withheld). | It was not playbook v2; clauses that read well but do not run remained targets for revision. |
| L2 | Opportunity specifications and best practices | The release index contained 11 best practices and 11 specifications; opportunity themes #5 and #9 remained observation items. | Countability did not mean that every theme had converged. |
| L3 | Executable plugins, skills, and recipes | The formal pack, including the `/sdd` path, skills, recipes, and adapters, was catalogued but largely stubbed on the main chain. | **A stub is not production-ready.** Main-chain hardening was Stage-2 work. |
| L4 | Maturity/effectiveness evidence pack | The handoff included evidence pointers and signal-type boundaries. | The signals were process observations and expert estimates, not platform-measured return; this paper does not reuse their numeric results. |

across twelve named themes: AI-ready story/living specification; UI prototyping; AI artifact review; cycle-time/AI-effectiveness measurement design; role handoff (observation); user harness/context pack; knowledge-base grounding; test engineering; token/model routing (observation); brownfield specification-driven development; agent nondeterminism evaluation; and CI toward AI control-system checks. Themes #5 and #9 retained observation status rather than being presented as fully converged, so management could count the handoff without erasing uncertainty within it.

**L3: scaffold, not production readiness.** The handoff contained a formal executable-asset pack whose redacted layout was: `commands/` (`/sdd` catalog), `skills/{intent,spec,design,verify,harness}/`, `recipes/` (ordered stage sequences), `adapters/` (coding-agent hosts), and `agents/` (sample wrappers only). The catalog existed, but the main execution chain was largely stubbed: many skill bodies were scaffolds, and stage recipes were not yet a hardened primary path. Org-A therefore recorded the assets as present but did not infer production readiness from the directory tree, and did not treat early agent wrappers as mature automation. Moving the primary chain from stub to ready was explicitly assigned to Consolidation.

The validation vehicle was a pilot change request on a live delivery track: the team exercised intent capture, living-specification update, increment application, and human-owned verification under retained quality gates. Acceptance for Explore stayed on whether that method chain could be run and cited, not on shipping the vehicle's full business scope. We do not report cycle-time or defect metrics for the vehicle here; those belong to the companion measurement manuscript.

**L4: bounded evidence.** The evidence layer contained maturity and effectiveness materials and stated their signal types. Workshop judgments were expert estimates and the materials were not presented as platform-measured return. For this path paper, L4 establishes that evidence boundaries were available to the gate; it is not used to introduce the companion paper's quantitative findings.

### *E. Phase-One Gate Decision*

At the early-August executive readout, management applied five questions: whether the playbook was citable; whether opportunity assets were countable; whether executable readiness was labeled ready or stub; whether effectiveness and maturity signals carried boundaries; and whether the next-stage path could be discussed concretely. The result was a pass for the Explore gate. The method had run end-to-end, the initial playbook and countable opportunity assets existed, and bounded signals could be reported.

The same review did *not* pass Scale admission. Replicability and seed coaches had not yet been demonstrated, the principal L3 execution chain remained largely stubbed, and the role, starter-kit, sponsor-cadence, and resource prerequisites were not all in place. This was not a contradictory verdict. G1 answered whether the organization had enough evidence to harden a feasible method; G2 would answer whether that hardened method could be transferred to another team.

The two-part decision is the practical value of the path. Without distinct gates, management would have had to label the pilot either a failure because it was not ready for broad diffusion, or a scale success because it completed its planned activities. The stage-gated account preserves the observed Explore achievement and the unresolved admission work at the same time.

### F. Consolidation Started; Scale Entry Remained Conditional

After the early-August closeout, Org-A entered Consolidation, with an indicative one-to-two-month window from early August through the end of September 2026. The active work was framed as upgrading playbook v1 toward a standardized v2 and hardening the L3 main chain. Seed coaches and "everyone can teach" were capability targets and the Stage-2 exit criterion; they were not reported as achieved at Explore closeout.

A consulting-side training and certification design also described Foundation, Practitioner, and Seed Coach tracks and a team-readiness certification concept. It was a cross-customer capability-preparation artifact at design-specification status: courseware, item bank, and learning platform implementation were pending. It therefore corroborates that scale enablement was being designed, but it is not evidence that Org-A executed training waves.

The resulting case posture is bounded but operationally useful. Layer 1 of the feasibility argument was in place because stages and gates had been defined. Layer 2 was observed because Org-A completed an end-to-end Explore cycle and passed G1. Layer 3 remained conditional because G2 had not been met: later Scale admission depended on the team-can-teach result, named owners, a hardened starter kit, and sponsor support. An indicative October calendar horizon was subordinate to those conditions. No new-team self-run, wave rollout, or organization-wide adoption was evidenced at the freeze point.

Org-A thus illustrates why movement from pilot evidence toward conditional Scale is a managed transition rather than a retrospective relabeling of pilot success. The case supplies evidence for feasibility, asset creation, and the start of hardening. It also supplies explicit negative evidence—stubs, missing transfer proof, and unmet prerequisites—that prevented an unsupported Scale claim. A later assessment can apply G2 to those same declared conditions without changing the meaning of the Explore result.

## V. Case Org-B

### A. Industrial Context and Evidence Basis

Org-B is an anonymized manufacturing enterprise. Unlike Org-A's multi-team agile setting, Org-B's incumbent product-development regime follows Integrated Product Development (IPD). The AI4SE path must therefore be adapted to IPD decision and lifecycle structure. We do not report a clause-level mapping of G1 or G2 onto Org-B's IPD decision points; that mapping is a design constraint for Explore, not an observed process rewrite. We do not claim that Org-B has completed an organization-wide IPD×AI4SE rewrite.

The evidence basis is an author briefing aligned to the statement of work (SOW). There is no consultant-side workspace mirror: client work remains on the client intranet. Facts in this section are therefore SOW- and briefing-level, and we mark delivery status explicitly.

### B. Adoption After Infeasible Alternatives

From about mid-July through mid-August 2026, other co-located consulting approaches led early exploration. Org-B concluded that those AI4SE transformation approaches were not feasible for its setting and selected the stage-gated path described here as the primary Explore method. We report this as a client decision outcome. We do not name the alternative providers, and we do not present a competitive scorecard.

From that transition through late August 2026, this team has been guiding multiple pilot teams through product research-and-development process analysis under the path. The calendar Explore window is mid-July to mid-September 2026. As of the manuscript freeze (late August 2026), Explore remains in progress. We do not write Org-B as Explore-closed, Consolidation-complete, or organization-wide scaled.

### C. Multi-Mode Pilots under an IPD Constraint

The Explore wave intentionally samples heterogeneous software modes rather than a single representative team embedded software; internal-use application development; large, complex enterprise business systems; and AI-agent-oriented development. Mode diversity is a sampling design for this Explore wave, intended to exercise heterogeneous delivery constraints inside one organization. It is not a completed mode-by-mode outcome set, and it is not evidence of enterprise rollout or of a multi-enterprise sample.

### D. Harness Intent, Scope, and Vignette Bound

The intended user-harness layer is SDD-oriented and non-exclusive: it should interoperate with Spec-Driven Development stacks of the OpenSpec / Superpowers / Spec Kit *class* (illustrative, not an allow-list) and remain extensible to peers, with MCP/CLI hooks into Org-B's internal toolchain. We claim neither production-complete integration nor single-stack lock-in.

The SOW mixes Explore work with later-stage preparation: Goals 1–2 (method / harness toolkit and end-to-end playbook) are Explore targets still open at freeze; Goals 3–4 (training/certification and seed coaches) are scale-preparation items, not observed Consolidation/Scale outcomes; Goal 5 is measurement-scheme design only (no Org-B numbers here).

As a recognition vignette, Org-B shows only that an IPD manufacturer can record a local feasibility decision and start a multi-mode Explore under harness-portability constraints. It does not show that alternatives failed in general, that Explore closed, that deliverables were accepted, or that Consolidation or organization-wide scale has started. Concrete negative process evidence at freeze: G1/G2 host reviews inside IPD are not yet mapped; that absence is an Explore design debt, not a completed IPD×AI4SE rewrite.

## VI. Discussion

### A. Reading the Two Cases Together

Org-A and Org-B occupy different positions on the same path (Table III), with unequal evidence depth. Org-A is the primary case: Explore completed, Consolidation started, Scale

TABLE III
ORG-A AND ORG-B ON THE SAME PATH

| | Org-A | Org-B |
|---|---|---|
| Calendar | ∼6-week Explore finished; Consolidation started | ∼2-month Explore, mid-July–mid-September; in progress |
| Process | Multi-team agile; retained quality gates | IPD incumbent; path must adapt |
| Pilot shape | Enablement plus validation vehicle | Multi-mode team sample in one wave |
| Adoption | Sponsored enablement; G1 pass / G2 hold | Method switch after infeasible alternatives |
| Harness | Playbook and plugin pack in handoff; main chain stubbed | Extensible multi-SDD harness plus MCP/CLI; not locked to one stack |
| Evidence depth | In-repo ledger and gate packet | Author briefing; no intranet mirror |

admission withheld, with an in-repo ledger and gate packet. Org-B is a secondary recognition vignette: the path was selected after alternatives failed a local feasibility test, and Explore remains open under IPD. The shared path vocabulary can be named in more than one process heritage; that does not make Org-B a second completed case.

The contrast also clarifies what "method recognition" is allowed to mean. Org-B's switch is a decision fact about local feasibility, not a proof that the path dominates unnamed alternatives in general, and not a claim that Org-B has already produced Org-A's L1–L4 ledger. IPD remains a stated incumbent constraint: this paper does not exhibit which IPD reviews will host G1 or G2 at Org-B, and it does not invent mid-September Explore closeout status.

Generic Stage-Gate practice already separates phases with business-case reviews [10]. The incremental experience recorded here is narrower: an AI4SE admission contract that treats a stubbed executable chain as Consolidation work, publishes a withheld Scale decision beside a passed Explore gate, and treats a second organization's still-open Explore as recognition rather than as a completed case. We do not claim a new general stage-gate theory.

### B. Manager Takeaways (Actionable)

1) **Next week (50+ engineer org).** Pick one delivery stream; freeze paradigm $p$ and harness $\mathcal{H}^p$ (for Org-A, SDD-oriented); ban tool-only rollouts that skip a citable method and a readiness-labeled ledger.
2) **At the first executive readout.** Put G1 and G2 on separate agenda lines. Ask the five Explore questions (citable method; countable opportunity assets; stub/ready labels; bounded signals; discussable Consolidation path) before any Scale calendar is debated.
3) **Playbook v1 clauses that flip first (Org-A lesson).** (i) guidance that "reads right" but has no runnable L3 path; (ii) promoting stub skills into autonomous agents; (iii) letting the apply path self-certify verify. Rewrite those before opening a second team.
4) **Before any Scale wave.** Name intent/spec/orchestration/verification owners, designate seed coaches, and ship a starter kit another team can run; keep unmet G2 items on a public hold list.
5) **Keep negative evidence visible.** An unmet Scale gate, an in-progress Explore, and unaccepted deliverables are decision inputs—not narrative failures to hide.

### C. Boundary with the Measurement Companion

This paper uses bounded effectiveness and maturity materials only as qualitative gate inputs (Org-A L4; Org-B Goal 5 as scheme design). The companion measurement paper's unit of analysis is a dual-lens reporting scheme—labeled capability and delivery signals at Org-A—not the stage-gated path. The two manuscripts share a short anonymized Org-A context and the same enablement window; they do not share main figures, and we do not import that paper's tables or treat its workshop or maturity numbers as results of the path.

## VII. THREATS TO VALIDITY

**External validity.** The primary longitudinal case is a single organization (Org-A). Org-B is an in-progress Explore at manuscript freeze, not a balanced second site with a matching gate packet. Mode diversity inside Org-B improves within-organization coverage; it does not create a multi-enterprise sample.

**Construct validity.** "Scale-ready" is a gate-and-asset construct in this paper, not a platform-measured return construct. Absence of completed rollout must not be read as proof that the method is ineffective, and bounded Explore signals must not be read as Scale completion.

**Internal validity / observer bias.** Authors facilitated enablement, gate framing, and asset authoring at Org-A, and they are guiding Org-B's Explore. Consolidation and readiness judgments may reflect consultant–client alignment rather than independent audit. We mitigate by binding factual sentences to a frozen evidence pack, by publishing negative evidence (stubbed L3; unmet G2; Org-B still in Explore), and by recording gate dispositions as client management decisions (Advance / Hold) rather than as consultant self-scores.

**Org-B verification limits.** Org-B prose relies on an SOW-aligned author briefing. The absence of an intranet artifact mirror limits external checking of those claims. The adoption-after-alternatives account is given at sector and pattern level without naming providers, so readers cannot independently reconcile competitive context.

**Conclusion validity.** A positive G1 plus a negative G2 is an internally consistent pair, not a contradictory evaluation. Treating either gate as a completed-scale result, or treating Org-B's in-progress Explore as a second completed case, would be an invalid conclusion.

## VIII. RELATED WORK AND CONCLUSION

### A. Related Work

**Methodological positioning.** We follow software-engineering case-study guidance on context, chain of evidence, and threats [2], and we treat the stage-gated path as a designed enablement artifact [3] illustrated primarily at

Org-A. We do not claim theoretical saturation or a completed design-science evaluate cycle with field-measured Scale outcomes.

**Productivity caution, not a measurement contribution.** Forsgren et al. argue that developer productivity is multi-dimensional and that single metrics mislead [1]. We take that caution as a reason not to treat a pilot slogan as Scale authorization. The industrial dual-lens reporting protocol that labels and jointly publishes capability and delivery signals is the companion paper's contribution, not ours.

**Sensemaking and classical SE layering as path rationale.** Cynefin decision domains motivate treating early AI4SE work as Complex probes rather than Clear rollouts [4]. Classical layered software engineering [7] motivates keeping process/methods/tools structure while upgrading executors and placing harness/SDD/agentic assets primarily in Methods. Fowler's harness writing supplies the bounded-context intuition we adapt for a user-harness certainty boundary [9]. None of these sources is re-evaluated here; they justify staging and placement vocabulary only.

**Industry retreat synthesis (rationale taxonomy only).** Thoughtworks' Utah and Europe Future of Software Engineering retreats map practitioner fault lines—rigor migration into specs/tests/constraints, a supervisory middle loop, verification scarcity, harness ownership, and decision-time bottlenecks [5], [6]. We borrow only that *fault-line taxonomy* as staging rationale (why Explore evidence, Consolidation hardening, and conditional Scale admission are distinct decisions). We do not import retreat anecdotes as Org-A/Org-B facts, do not treat retreat consensus as multi-site validation of our gates, and do not claim the retreats prescribe our five-question packet; SPACE remains our productivity caution [1].

**Spec-driven and agentic practice.** Gupta's preprint discusses spec-driven development together with agentic frameworks [8]. Our use of SDD and Agentic Engineering is narrower: they are two directions inside a managed path, with harness and operating-model obligations attached to later stages.

**Maturity models for AI-assisted SE.** Tereci et al. propose a conceptual maturity model and research agenda for AI-assisted software development [11]. Anderson's experience-report preprint frames progression from assisted coding toward more autonomous systems [12]. Those works target organizational or codebase maturity theory. Our unit of analysis is a stage-gated transformation path and the admission contract of its gates, not a new maturity model.

**Differentiation from generic stage-gate.** Classical Stage-Gate product processes separate discovery, development, and launch with business-case gates [10]. Our path reuses the *idea* of evidence-based admission, but the gate objects differ: AI4SE admission inspects a citable method, a stub-versus-ready executable ledger, and transfer preconditions—not a product business case. Generic Stage-Gate does not, by itself, prevent equating a coding-assistant pilot with organization-wide diffusion, nor does it encode stub $\neq$ ready for agent skill packs. Relative to AI-assisted maturity models, and to that generic practice, this paper records an AI4SE-specific pairing: a withheld Scale admission beside a completed Explore at Org-A, plus a thinner Org-B recognition vignette rather than a second completed asset ledger. We do not claim that pairing is unique in the literature; it is the experience we can evidence.

### B. Conclusion

We asked how an industrial AI4SE path from pilot toward scale can be designed (Q1), what Org-A's Explore produced in support of Consolidation and of only-conditional Scale (Q2), and what a still-open Org-B Explore can legitimately show about adoption after infeasible alternatives (Q3).

The transferable artifact is the stage-gated path: Explore, Consolidation, and conditional Scale, operated as evidence reviews at G1 and G2. Org-A shows that a six-week Explore can yield a citable method, countable opportunity knowledge, and a stub-versus-ready ledger, and that those results can justify Consolidation without authorizing Scale. Org-B contributes only a decision fact and an in-progress multi-mode Explore under IPD; it does not contribute a second completed gate packet. Neither organization is claimed as organization-wide scale; neither reports platform-measured return.

Future camera-ready updates may refresh Org-B's mid-September delivery status once author-confirmed. They should not convert an unmet G2, or an in-progress Explore, into a completed-scale claim.

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## Data Availability

Anonymized, self-contained evidence for this manuscript is released at https://github.com/ai4se-works/artifacts/tree/main/manuscripts/stage-gated-ai4se-path/data (CC-BY-4.0). The pack includes a gate rubric (G1/G2), a public L1 method card (chapter map, principles, artifact chain, process gates, paraphrased normative clauses, and declared v1 gaps), an L1–L4 asset ledger, a twelve-theme index with observation flags, a redacted L3 stub map, and the Org-A timeline used in the case narrative. It is intended for inspection of those claims without access to partner intranets. The full client-facing playbook text, opportunity Spec/BP bodies, plugin sources, unredacted change records, and Org-B workspace content remain confidential. Quantitative maturity and workshop tables are not reproduced here; they appear in the companion measurement data pack in the same artifacts repository.

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